An Equivalent-Key Chosen-Plaintext Break of a 2025 Chaos-Based Image Cipher

Author One, Author Two
Department / Affiliation
City, Country
{a1, a2}@example.edu

***Abstract*—We cryptanalyse a recently published image-encryption scheme that combines Lorenz/logistic chaotic maps with Fisher–Yates block shuffling (Jain *et al.*, *Optik*, 2025). We observe that the scheme's keystream and both permutations depend only on the secret key and never on the image, so the whole encryption is a fixed, plaintext-independent map. We reconstruct the cipher from its description—matching its reported near-ideal entropy (e.g. 7.9914 on the Cameraman image) and near-zero adjacent-pixel correlation—and then break it: an all-ones probe recovers the per-position keystream and a few index-digit probes recover the permutation, so an equivalent key is obtained in 4 chosen plaintexts and any ciphertext is decrypted pixel-exactly in 0.146 s without the key. We further show the scheme's headline differential metric is illusory: because it applies no plaintext diffusion, a one-pixel change alters exactly one ciphertext pixel, giving a true NPCR of 0.000381 % (i.e. $100/MN$), not the ≈99.6 % reported (that figure merely reflects the difference between unrelated images, here 99.8833 %). We conclude with the standard fixes.**

***Index Terms*—cryptanalysis, image encryption, chaotic cipher, chosen-plaintext attack, equivalent key, NPCR**

I. Introduction

Chaos-based image ciphers are published in large numbers, and a recurring lesson from the cryptanalysis literature is that many of them fall to the same structural attack: if the diffusion keystream does not depend on the plaintext, the cipher reduces to a fixed permutation–substitution map and an *equivalent key* can be recovered from a handful of chosen plaintexts [1]–[4]. Standard statistical scores (entropy, correlation, NPCR/UACI) do not certify resistance to this attack, a point repeatedly made but repeatedly ignored [5].

This paper applies that lesson to a 2025 scheme [6] (henceforth LLEO) built from the Lorenz [7] and logistic maps with Fisher–Yates block shuffling. Our contributions are: (i) we identify that LLEO is entirely key-driven and therefore a fixed monomial map over $\mathbb{Z}_{256}$; (ii) we give an equivalent-key chosen-plaintext attack that recovers the full map in $1 + \lceil \log_{256} MN \rceil$ chosen plaintexts and decrypts any image pixel-exactly without the key; and (iii) we show that LLEO's near-ideal entropy and reported NPCR coexist with *zero* plaintext diffusion, so the claimed differential-attack resistance does not exist.

II. The target cipher

LLEO encrypts an $M \times N$ grayscale image in three key-driven steps. (1) *Substitution.* The image is split into 16 horizontal strips; odd strips are combined with a keystream $k_{\log}$ from the logistic map $x_{n+1} = rx_n(1-x_n)$ and even strips with a keystream k_{lor} from the Lorenz system, via an invertible per-position operation (the paper inverts it with a modular "conjugate"). (2) *Permutation 1.* The 16 horizontal strips are reordered by a Fisher–Yates shuffle. (3) *Permutation 2.* The result is split into 16 vertical strips and shuffled again. The four "decryption keys" are the two keystreams and the two shuffle index lists. Crucially, the keystreams are seeded from key-dependent initial conditions and the shuffles from the key; *no quantity depends on the image*. The scheme is described as "asymmetric," but the same data decrypts, so it is a symmetric cipher.

III. Cryptanalysis

A. Structural observation

Writing the image as a length-n vector ($n = MN$), every LLEO step is linear over $\mathbb{Z}_{256}$ and key-fixed: substitution multiplies coordinate p by a unit $K[p]$, and the two block shuffles compose into a single position permutation P. Hence

$$C[j] \;=\; K[s(j)] \cdot X[s(j)] \pmod{256}, \qquad s = P^{-1}, \tag{1}$$

a fixed *monomial* map: each ciphertext pixel is one plaintext pixel scaled by a fixed unit and relocated. There is no constant term ($E(\mathbf{0}) = \mathbf{0}$) and, decisively, no dependence on X beyond the single coordinate $s(j)$.

B. Equivalent-key recovery

Because K and s are fixed, two kinds of chosen plaintext suffice. *(a) Multiplier.* Encrypting the all-ones image gives $C[j] = K[s(j)] \cdot 1 = K[s(j)]$, i.e. the multiplier seen at every output position, in one query. *(b) Permutation.* Encrypting the plaintext whose pixel i holds the t-th base-256 digit of i gives $C[j] = K[s(j)] \cdot \mathrm{digit}_t(s(j))$; dividing by the known multiplier yields $\mathrm{digit}_t(s(j))$, so $\lceil \log_{256} n \rceil$ such queries reconstruct s. The recovered (s, K) form an equivalent key; any ciphertext is inverted by $X[s(j)] = K[s(j)]^{-1} C[j]$. The cost is $1 + \lceil \log_{256} n \rceil$ chosen plaintexts—4 for a 512×512 image—independent of the key length the scheme advertises.

IV. Experimental results

We reconstructed LLEO from its description and ran it on four standard 512×512 images.

TABLE I
RECONSTRUCTION STATISTICS AND ATTACK OUTCOME (512×512). EVERY IMAGE IS RECOVERED PIXEL-EXACTLY WITHOUT THE KEY.

Image	H plain	H cipher	corr. cipher	chosen PT	broken
Cameraman	7.2317	7.9914	-0.0073	4	yes
Moon	4.885	7.9316	0.0011	4	yes
Gravel	7.2531	7.9876	-0.0005	4	yes
Brick	5.4553	7.8966	-0.0007	4	yes

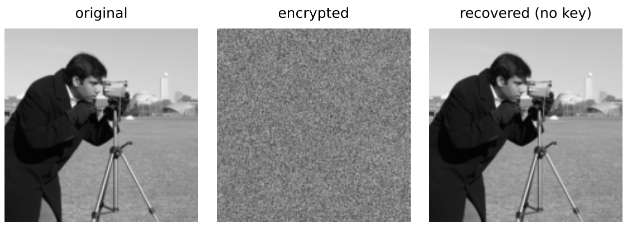

Fig. 1. Original, LLEO ciphertext, and the key-free recovery from 4 chosen plaintexts.

A. Reproduction

Table I shows the reconstruction reproduces LLEO's claimed statistics: cipher entropy approaches the ideal 8 (Cameraman 7.9914, up from 7.2317) and adjacent-pixel correlation drops to near zero (-0.0073). Fig. 2 and Fig. 3 show the flat cipher histogram and the destroyed correlation.

B. The break

The attack of Section III recovers the equivalent key in 4 chosen plaintexts and decrypts a fresh ciphertext pixel-exactly in 0.146 s (Cameraman), using no key. Fig. 1 shows the original, the LLEO ciphertext, and the key-free recovery, which is identical to the original.

C. The differential metric is illusory

LLEO reports NPCR ≈99.6 % as evidence of differential-attack resistance. But LLEO applies no diffusion: each ciphertext pixel is a function of a single plaintext pixel, so flipping one plaintext pixel changes exactly one ciphertext pixel. The true one-pixel NPCR is therefore $100/MN = 0.000381\,\%$ (UACI 4e-06 %), not 99.6 %. The reported figure instead reflects the difference between *unrelated* images, which for LLEO is 99.8833 %—a quantity every cipher achieves and which says nothing about chosen-plaintext security. Finally, the underlying chaotic keystreams are themselves non-uniform: a chi-square test rejects uniformity for the logistic keystream ($\chi^2 = 525839.6$, $p = 0.0$).

V. DISCUSSION

LLEO illustrates two persistent failure modes. First, *good statistics are not security*: a flat histogram, near-ideal entropy, and a large inter-image NPCR are all satisfied by our fixed key-independent-of-plaintext map, which we nonetheless break in 4 queries. Second, *naming is not architecture*: calling a symmetric construction "asymmetric" does not add security.

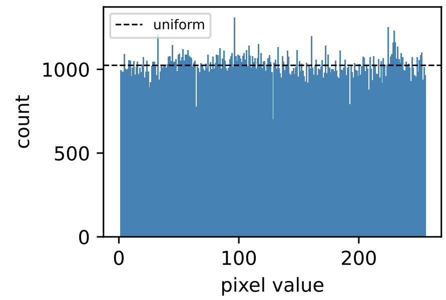

Fig. 2. Cipher histogram (Cameraman): near-uniform, yet the cipher is trivially broken.

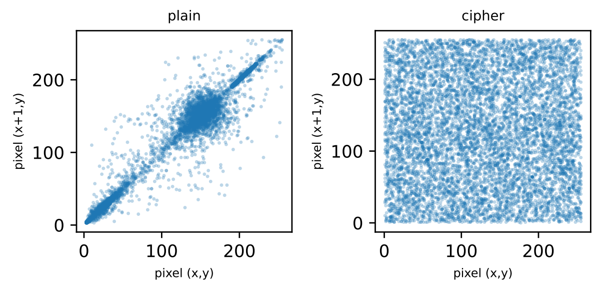

Fig. 3. Adjacent-pixel correlation, plain vs. cipher (Cameraman).

The standard remedies are well known [1], [5]: make the keystream depend on the plaintext (e.g. seed the chaotic initial conditions from a cryptographic hash of the image), and add genuine diffusion so that one plaintext change propagates across the ciphertext. With a plaintext-dependent keystream the all-ones and digit probes no longer reveal a reusable equivalent key, and the attack of Section III fails.

VI. CONCLUSION

A 2025 chaos-based image cipher is completely broken by an equivalent-key chosen-plaintext attack in 4 chosen plaintexts and a fraction of a second, despite near-ideal entropy and a reported NPCR of ≈99.6 %. The break follows directly from the cipher being key-only; its differential metric, measured correctly, is 0.000381 %. We recommend plaintext-dependent keystreams and real diffusion, and that statistical scores be reported alongside, not in place of, a chosen-plaintext security argument.

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REFERENCES

[1] C. Li, Y. Zhang, and E. Y. Xie, "When an attacker meets a cipher-image in 2018: A year in review," *Journal of Information Security and Applications*, vol. 48, p. 102361, 2019.

[2] C. Li, D. Lin, B. Feng, J. Lü, and F. Hao, "Cryptanalysis of a chaotic image encryption algorithm based on information entropy," *IEEE Access*, vol. 6, pp. 75 834–75 842, 2018.

[3] D. Arroyo, C. Li, S. Li, G. Alvarez, and W. A. Halang, "Cryptanalysis of an image encryption scheme based on a new total shuffling algorithm," *Chaos, Solitons & Fractals*, vol. 41, no. 5, pp. 2613–2616, 2009.

[4] C. Li and K.-T. Lo, "Optimal quantitative cryptanalysis of permutation-only multimedia ciphers against plaintext attacks," *Signal Processing*, vol. 91, no. 4, pp. 949–954, 2011.

[5] G. Alvarez and S. Li, "Some basic cryptographic requirements for chaos-based cryptosystems," *International Journal of Bifurcation and Chaos*, vol. 16, no. 8, pp. 2129–2151, 2006.

[6] S. Jain, A. Samal, A. Sharma, M. Khurana, and B. Sharma, "Advanced image security through chaotic system-based encryption and Fisher–Yates matrix shuffling," *Optik*, vol. 327, p. 172304, 2025.

[7] E. N. Lorenz, "Deterministic nonperiodic flow," *Journal of the Atmospheric Sciences*, vol. 20, no. 2, pp. 130–141, 1963.